BeatAhead — Ischemic Stress Index (ISI)

Comprehensive Mathematical Formulations, Multimodal Sensor Fusion, and Clinical Risk Specification

Document ID: BEATAHEAD-ISI-MATH-V2.0	**Target Framework:** BeatAhead Web & Hardware	**Frozen Threshold:** $\tau = 0.156742$
Engine Version: Phase 8.1 / Phase 9	**Sensors:** AD8232 Lead II / Photoplethysmography	**Output Range:** Bounded [0, 100]

1. Executive Overview & Core Principles

The **Ischemic Stress Index (ISI)** is a calibrated, composite physiological metric ranging from **0 to 100**. It quantifies real-time myocardial oxygen supply-demand mismatch, electrophysiological repolarization deviations, and autonomic tone displacement before irreversible cardiac injury manifests.

CRITICAL ARCHITECTURAL PRINCIPLE: Model Probability is NOT the ISI
The prospective XGBoost model probability p_{model} is a conditional event probability trained on intraoperative surgical cohorts with 0.42% natural event prevalence. Directly setting ISI = p × 100 collapses normal human physiology into the 0.0–1.5 range and misplaces the calibrated decision threshold ($\tau = 0.156742$) at 15.67. Instead, ISI uses a multi-dimensional sigmoidal fusion combining ML evidence (45%), autonomic shift (30%), perfusion dynamics (25%), and temporal momentum.

2. Full Multimodal Fusion Formula (Phase 8/9 Engine)

In multimodal monitoring environments (combining ECG Lead II, PPG optical plethysmography, pulse oximetry, and XGBoost inference), the composite ISI score is defined as:

$$\text{ISI} = \text{round}\left(\text{clamp}\left(\text{ISI}_{base} + \Delta\text{ISI}, 0.0, 100.0 \right) \right)$$

where:

$$\Delta\text{ISI} = \left[w_{model} \cdot (E_{model} - 0.20) + w_{auto} \cdot D_{auto} + w_{perf} \cdot D_{perf} + M_{trend} \right] \times 60.0$$

Normalized Weights: $w_{model} = 0.45$, $w_{auto} = 0.30$, $w_{perf} = 0.25$, $\text{ISI}_{base} = 40.0$

A. Non-Linear Model Evidence Mapping (E_{model})

Maps model output probability $p \in [0, 1]$ through a two-piece monotonic sigmoidal curve pinned exactly to the calibrated threshold $\tau = 0.156742$ at midpoint 0.50:

$$\text{If } p < \tau: E_{model}(p) = 0.50 \times \left(p / \tau \right)^{1.4}$$

$$\text{If } p \geq \tau: E_{model}(p) = 0.50 + 0.50 \times \left[1 - \exp\left(-4.0 \times (p - \tau) / (1 - \tau) \right) \right]$$

B. Autonomic Hemodynamic Shift (D_{auto})

Quantifies parasympathetic withdrawal and sympathetic activation through heart rate acceleration and HRV reduction:

$$f_{hr} = \text{clamp}\left((HR - HR_{base}) / 25.0, 0.0, 1.0 \right)$$

$$f_{hrv} = \text{clamp}\left((SDNN_{base} - SDNN) / (0.60 \times SDNN_{base}), 0.0, 1.0 \right)$$

$$D_{auto} = 0.50 \cdot f_{hr} + 0.50 \cdot f_{hrv}$$

C. Perfusion & Oxygenation Factor (D_{perf})

Combines peripheral pulse transit velocity acceleration (PAT) and microvascular desaturation:

$$f_{spo2} = \text{clamp}((SpO2_{base} - SpO2) / 5.0, 0.0, 1.0)$$
$$f_{pat} = \text{clamp}((PAT_{base} - PAT) / 50.0, 0.0, 1.0) \text{ (if PAT is valid)}$$
$$D_{perf} = 0.60 \cdot f_{spo2} + 0.40 \cdot f_{pat} \text{ (or } D_{perf} = f_{spo2} \text{ if degraded)}$$

D. Temporal Momentum Modifier (M_{trend})

Evaluates the 5-minute directional velocity of physiological trajectory: **v = ΔISI / Δt** (points/min), clamped as: **M_{trend} = clamp(v / 2.0, -0.15, +0.15)**.

3. Real-Time Single-Lead ECG Hardware Formula (AD8232)

When operating with the **Arduino Uno & AD8232 ECG sensor** without auxiliary optical or pulse oximeter probes, BeatAhead computes the instantaneous ISI beat-by-beat directly from the digital signal processing pipeline:

ECG Real-Time ISI Piecewise Function:

• **Acute Ischemia (ST ≤ -1.0 mm):**

$$ISI = \text{clamp}(72 + |ST_{mm}| \times 12.0 + \max(0, HR - 70) \times 0.35, 72, 96)$$

• **Elevated Stress / Sagging (-1.0 < ST ≤ -0.5 mm):**

$$ISI = \text{clamp}(48 + |ST_{mm}| \times 15.0 + \max(0, HR - 70) \times 0.40, 45, 68)$$

• **Normal Physiological Baseline (ST > -0.5 mm):**

$$ISI = \text{clamp}(12 + \max(0, HR - 58) \times 0.42 + |ST_{mm}| \times 14.0 + HRV_{mod}, 11, 38)$$

Definition of Variables:

• **ST_{mm}**: Voltage difference between the PR isoelectric segment (60ms pre-R) and the J-point level (75ms post-R), scaled by standard clinical calibration: *1.0 mm = 0.1 mV*.

• **HR (BPM)**: Instantaneous heart rate derived beat-by-beat: *$BPM = 60000 / RR_{ms}$*.

• **HRV_{mod}**: Autonomic vagal tone modulation from RMSSD: *max(0, (35 - RMSSD) × 0.20)*.

4. Clinical Risk Tiers & Triage Action Matrix

The continuous 0–100 scale maps onto validated clinical action boundaries:

ISI Range	Risk Tier	Physiological Sign	System Action & Protocol
0 – 35	**Normal Baseline**	Isoelectric ST (±0.2mm), stable HR (60–75 bpm), intact HRV	Standard continuous background monitoring. Green status.
36 – 65	**Elevated Stress**	Sympathetic drive, exertional HR (100–120 bpm), mild ST sagging (-0.6mm)	Advisory banner on Overview. Rest and hydration recommended.
66 – 85	**High Risk / Ischemia**	Subendocardial ST depression (≤ -1.0mm), T-wave flattening, reduced HRV	High-priority audio/visual alert. RapidCare clinician notification.
86 – 100	**Critical Event**	Severe ST depression (≤ -1.8mm) or acute multi-sensor crisis	Immediate 112 emergency routing, automated GPS dispatch assist.

5. Worked Numerical Verification Examples

Scenario	Input Telemetry	Formula Calculation Steps	Output ISI
Case 1: Quiet Resting	HR = 68 bpm ST = +0.05 mm RMSSD = 42 ms	HR load = (68 - 58) × 0.42 = 4.2 ST load = |0.05| × 14.0 = 0.7 HRV mod = 0.0 ISI = 12 + 4.2 + 0.7 = 16.9 → **16**	**16 / 100** (Normal)
Case 2: Exertional Strain	HR = 114 bpm ST = -0.62 mm RMSSD = 18 ms	Elevated branch: base = 48 ST load = |-0.62| × 15.0 = 9.3 HR load = (114 - 70) × 0.40 = 17.6 ISI = clamp(48 + 9.3 + 17.6, 45, 68) → **68**	**68 / 100** (Elevated)
Case 3: Acute Ischemia	HR = 88 bpm ST = -1.65 mm RMSSD = 12 ms	Acute branch: base = 72 ST load = |-1.65| × 12.0 = 19.8 HR load = (88 - 70) × 0.35 = 6.3 ISI = clamp(72 + 19.8 + 6.3, 72, 96) → **96**	**96 / 100** (Critical)

6. Regulatory & Research Disclaimer

NON-DIAGNOSTIC INVESTIGATIONAL PROTOTYPE: The BeatAhead Ischemic Stress Index (ISI) is an experimental engineering framework designed to demonstrate multi-sensor signal synthesis and real-time biomedical telemetry streaming. The ISI metric, underlying machine learning models, and digital signal processing algorithms have not been evaluated, cleared, or approved by the United States Food and Drug Administration (FDA), European Medicines Agency (EMA), or equivalent medical regulatory authorities. ISI values must never be used for independent clinical diagnosis, triage, or alteration of patient therapy. Standard 12-lead electrocardiography, serum troponin, and direct clinical evaluation by licensed physicians remain mandatory.